

UNIVERSITY DEPARTMENT NETWORK IMPLEMENTATION

Department of Computer Science & Software Engineering

Comprehensive Project Report: University Multi-Department Network

Inter-VLAN Routing, Dynamic Host Configuration, and Infrastructure Wireless Architecture using Cisco Packet Tracer

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1. Executive Summary & Design Overview

This technical documentation establishes a fully functional and scalable multi-department enterprise campus network architecture developed inside Cisco Packet Tracer. The network is specifically tailored for a modern university topology containing specialized zones: the **Information Technology (IT) Department**, the **Computer Science (CS) Department**, and a centralized **University Server Infrastructure**.

To preserve network efficiency, minimize broad broadcast domains, and guarantee rigorous logical access control across departments, the network incorporates Virtual Local Area Networks (VLANs). Seamless multi-subnet routing is successfully handled via Router-on-a-Stick sub-interface routing configurations. Furthermore, dynamic parameters (IP addresses, default gateways, and custom DNS options) are automatically propagated via dynamic address ranges, eliminating manual configuration headaches on client workstations.

1.1 Network Architecture Objectives

- **Logical Traffic Segregation:** Implementing standalone VLAN identifiers to isolate sensitive administrative, infrastructure, and generic computer labs.
- **Scalable Inter-VLAN Communication:** Utilizing a central high-performance router to securely bridge multi-subnet endpoints.
- **Infrastructure Wireless Enterprise Deployment:** Provisioning uninterrupted IEEE 802.11 dynamic access points allowing portable client nodes to instantly connect to internal campus services.
- **Centralized Server Resources:** Deploying web and automatic IP provisioning platforms to host academic data portals.

2. Logical Subnetting and VLAN Partitioning

The physical switching infrastructure is split down into discrete logical channels to streamline tracking and maintain tight security. Below is the precise architectural mapping utilized for provisioning the Layer-2 switches and the central Layer-3 gateway:

VLAN ID	Department Name	Subnet Range	Gateway IP	Assigned Interface Pools
VLAN 10	Computer Science (CS)	192.168.10.0 / 24	192.168.10.254	Fa0/1, Fa0/2, Fa0/6
VLAN 20	Information Technology (IT)	192.168.20.0 / 24	192.168.20.254	Fa0/3, Fa0/4

2.1 Core Layer-2 Switch Port Profiles

The core layer-2 device named MAIN_SWITCH contains dedicated access boundaries to completely isolate specific departments. FastEthernet interfaces are explicitly hardcoded to drop multi-vlan frame leaking, while the uplink port leading directly to the router is designated as an 802.1Q trunk.

```
MAIN_SWITCH# show running-config (Interface Access Matrix)
```

```
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/2  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/3  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/4  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/6  
  switchport access vlan 10  
  switchport mode access  
!
```

3. Layer-3 Gateway & Dynamic Pooling (DHCP) Engine

To ensure automatic setup across hundreds of student endpoints, the central equipment running under the hostname VLAN_ROUTER is loaded with a dynamic allocation mechanism. Explicit address exclusions are defined to shield critical static points—such as the default gateways themselves—from address duplication bugs.

```
VLAN_ROUTER# show running-config (DHCP & Sub-interface Setup)

!
hostname Router
!
ip dhcp excluded-address 192.168.10.254
ip dhcp excluded-address 192.168.20.254
!
ip dhcp pool VLAN10
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.254
  dns-server 8.8.8.8
!
ip dhcp pool VLAN20
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.254
  dns-server 8.8.8.8
!
```

3.1 Central Server Pool Properties

The centralized server environment UNIVERSITY_SERVER features customized pools configured to serve endpoints over FastEthernet interfaces. This guarantees that whenever a student machine boots up anywhere inside the campus, it seamlessly requests and receives a certified token.

System Panel: UNIVERSITY_SERVER — Centralized DHCP Pooling Properties

Interface: FastEthernet0 | Service Status: Off (Handled by Router)

Pool Name	Default Gateway	Start IP Address	Subnet Mask	Max Users
serverPool	0.0.0.0	192.168.10.0	255.255.255.0	512

4. Wireless Network Infrastructure & Access Integration

To accommodate portable computing hardware including laptop nodes (Laptop2 and Laptop3), specialized infrastructure access points are operational on the campus. Wireless interface tracking monitors active client link properties. Laptops utilize dedicated WPC300N expansion modules to communicate smoothly with the distribution tier.

Hardware Profile & State: Laptop2 / Laptop3 Wireless Monitor

Infrastructure Mode: Dynamic Connection Active

Status: You have successfully connected to the access point

Signal Strength: 100% | Link Quality: 100%

Wireless Adapter Profile: Active Link | Standard: 2.4 GHz (Wireless-N Notebook Adapter)

5. Connectivity Verification & ICMP Diagnostics

To validate that data packets can safely travel across logical department borders, extensive ICMP (ping) operations were performed between different subnets. During the initial startup phase, the first packet might trigger an ARP lookup causing a single timeout. Once resolved, the communication achieves perfect transmission rates.

5.1 Diagnostics Session from CS Subnet (CS_VLAN_PC1) to IT Subnet (192.168.20.2)

```
CS_VLAN_PC1 > command prompt > ping 192.168.20.2
```

```
C:\>ping 192.168.20.2
```

```
Pinging 192.168.20.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.2: bytes=32 time=4ms TTL=127
```

```
Ping statistics for 192.168.20.2:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
C:\>ping 192.168.20.2
```

```
Pinging 192.168.20.2 with 32 bytes of data:
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.20.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

5.2 Diagnostics Session from IT Subnet (IT_VLAN_PC2) to CS Subnet (192.168.10.2)

```
IT_VLAN_PC2 > command prompt > ping 192.168.10.2
```

```
C:\>ping 192.168.10.2
```

```
Pinging 192.168.10.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

6. Hosted Campus Application & HTTP Services

The UNIVERSITY_SERVER node acts as a core data center module by activating HTTP/HTTPS engines. This allows student endpoints to parse intranet HTML domains natively across the network.

Service Panel: UNIVERSITY_SERVER — Active Application Hosting Services

HTTP Service: On | HTTPS Service: On

Internal File System Manager Content:

- copyrights.html (edit/delete)
- cscoptlogo177x111.jpg (delete)
- helloworld.html (edit/delete)
- image.html (edit/delete)
- index.html (edit/delete)

7. Lab Conclusion & Implementation Summary

The project successfully satisfies all design requirements of the University Computer Network implementation. By partitioning the network environments into VLAN 10 (CS Department) and VLAN 20 (IT Department), broadcast storm issues are tightly mitigated.

Dynamic routing configurations via sub-interfaces maintain communication fluidity while the system achieves stable 0% packet loss metrics upon environment convergence. The design stands ready for subsequent institutional expansion parameters.